

RELTWIN: CONTRASTING LOCALLY VALID TIMESTAMP ANSWERS FOR QUERY-SPECIFIC AUDIO GROUNDING

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ABSTRACT

Audio temporal grounding must distinguish which event occurrence answers a query, even when opposite event orders coexist in one recording. Models can instead return the same window for inverse queries, revealing a query-window binding error. We introduce RelTwin, which constructs opposite-order windows from shared audio excerpts and contrasts their timestamp answers under each query. Both windows have real acoustic support; the requested order determines the correct answer. This explicit answer-level competition complements sequence supervision and preserves direct timestamp generation at inference. Compared with paired SFT, RelTwin improves joint localization by 31.25 points on the synthetic relation-development set and retains an advantage under timing intervention. Across three training seeds, it averages 59.16 mIoU on SpotSound-Bench, surpassing the prior main-table best by 1.26 points.

Index Terms— Audio temporal grounding, timestamp answer contrasts, query-window binding, paired supervision

1. INTRODUCTION

Audio temporal grounding maps a query to event intervals [1, 2]. In a recording containing both “dog then rooster” and “rooster then dog,” the descriptions refer to different windows. Recognizing the categories establishes presence; answering the query requires selecting the requested order. Returning the same real window for both queries is a *query-window binding error*. We target this distinction between event detection and query-specific localization.

RelTwin (Fig. 1) addresses this binding problem through *locally valid timestamp contrasts*. Shared excerpts form opposite-order windows in one recording. Each window contains a real event sequence, but its correctness as an answer depends on the query. Sequence supervision rewards correct timestamp answers individually; the added candidate loss explicitly compares the two answers under each query. This local competition trains query-window binding while inference retains direct timestamp generation.

We make three contributions. First, we characterize query-window binding errors when inverse relations co-occur in one recording. Second, we introduce locally valid timestamp contrasts to supervise the query-specific choice between competing answers. Third, paired diagnosis and a timing intervention demonstrate improved query-window binding, alongside gains on SpotSound-Bench.

2. RELATED WORK

Audio temporal grounding. Weakly supervised grounding learns temporal correspondences from clip-level captions, reducing dependence on frame-level annotations [3]. Audio-language models such as Pengi [4], Qwen2-Audio [5], and Audio Flamingo 3 [6] provide audio-conditioned text generation. For temporal localization, TimeAudio introduces temporal markers and absolute time-aware encoding [7], while SpotSound interleaves timestamps with audio tokens and uses absent-event queries to suppress hallucinated intervals [8]. SpotSound thus combines event-presence supervision with timestamp prediction. RelTwin targets query-specific selection when inverse relations coexist: both descriptions have acoustic support in the recording, and the requested relation determines which occurrence is the answer.

Compositional audio-language learning. CLAP aligns audio and text in a shared representation space [9]. T-CLAP introduces order-reversed descriptions and a temporal-focused contrastive objective [10]. CompA evaluates event order and attribute binding, and trains CompA-CLAP with composition-aware hard negatives and modular contrastive learning [11]. CoSTALA combines spatio-temporal contrast with local event-caption alignment and feature consistency [12]. These methods organize audio-text representations around compositional relations, including their local structure. RelTwin places the supervision on complete timestamp answer sequences: their query-conditioned likelihoods determine the preference between competing windows. This trains relational selection in the output space of a generative grounding model.

Hard negatives for temporal grounding. In compositional video grounding, SHINE combines generated hard-

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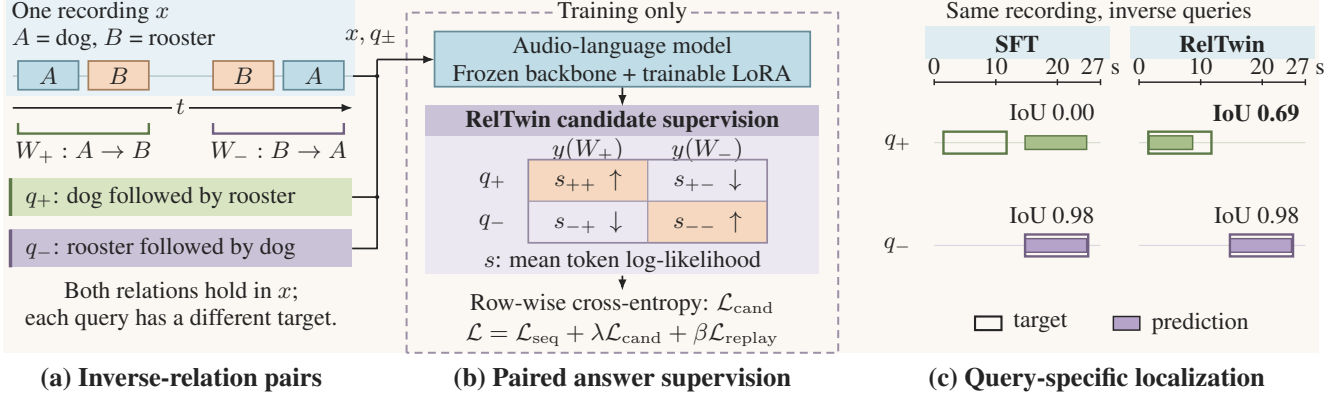


Fig. 1. RelTwin binds inverse queries to different windows in one recording. (a) Reused excerpts form two local targets. (b) The dashed enclosure marks training-only computation; shaded diagonal cells identify correct timestamp answers, and up/down arrows indicate training preferences. (c) In a relation-development example, SFT repeats one answer while RelTwin selects the corresponding window for each query. Outlines mark targets; filled bars show predictions, evaluated by set-IoU.

negative queries with within-video and cross-query saliency ranking [13]. Its training compares segment relevance as well as query compositions. RelTwin makes complete timestamp answers the candidates for comparison within one recording. Each candidate denotes an entire event sequence with real acoustic support. Under a fixed query, one answer matches the requested order and the other answers the inverse relation; reversing the query exchanges their labels while preserving the audio context. Comparing the length-normalized sequence log-likelihoods explicitly binds each relation query to its corresponding local answer.

3. QUERY-SPECIFIC WINDOW LEARNING

3.1. Two true relations, two different answers

In one recording x , RelTwin arranges the same ESC-50 excerpts [14], A and B , as $A \rightarrow B$ and $B \rightarrow A$. Their local targets W_+ and W_- run from the first excerpt’s onset to the second’s offset, including the gap. Queries q_+ and q_- request the respective orders. Both windows have acoustic support, but W_- is positive for q_- and negative for q_+ ; reversing the query exchanges their labels.

The templates “followed by” and “and then” yield two inverse-query pairs per recording. Earlier-order layouts alternate (Section 5.2). Category rehearsal targets the union of both occurrences of A or B , retaining category-grounding supervision.

3.2. Contrast timestamp answers during training

We serialize an interval set S as comma-separated spans, each formatted like from 1.500s to 11.750s (three-decimal precision). Its supervised assistant-completion to-

kens $y(S)$ exclude the prompt and are scored by

$$s_\theta(x, q, S) = \frac{1}{|y(S)|} \sum_t \log P_\theta(y_t | x, q, y_{<t}). \quad (1)$$

The 2×2 matrix M , with entries $M_{ij} = s_\theta(x, q_i, W_j)$ for $i, j \in \{+, -\}$, indexes queries by rows and answers by columns. Its diagonal labels $Y = I_2$ identify the correct query-answer pairs (Fig. 1b). Sequence supervision minimizes $\mathcal{L}_{\text{seq}} = -(M_{++} + M_{--})/2$, rewarding the two correct responses individually. It does not explicitly compare them with the other locally valid window. RelTwin trains the relative preferences $M_{++} > M_{+-}$ and $M_{--} > M_{-+}$ through row-wise probabilities $p_i(j) = \text{softmax}_j(M_{ij}/\tau)$, where $\tau > 0$. The candidate loss is

$$\mathcal{L}_{\text{cand}} = -\frac{1}{2}[\log p_+(+) + \log p_-(-)]. \quad (2)$$

For a separately sampled category example (x_r, q_r, S_r) , the rehearsal loss is $\mathcal{L}_{\text{replay}} = -s_\theta(x_r, q_r, S_r)$. Together,

$$\mathcal{L} = \mathcal{L}_{\text{seq}} + \lambda \mathcal{L}_{\text{cand}} + \beta \mathcal{L}_{\text{replay}}. \quad (3)$$

Paired SFT sets $\lambda = 0$; RelTwin uses $\lambda > 0$ with identical relation and rehearsal data. Both directly generate timestamps at inference, without candidates or an additional scorer.

3.3. Joint localization and relation selection

For interval unions, $\text{IoU}_{\text{set}}(S, G) = |S \cap G|/|S \cup G|$ (zero for an empty union). Given predictions $\widehat{W}_i$, define $u_{ij} = \text{IoU}_{\text{set}}(\widehat{W}_i, W_j)$. PairAcc requires $\min(u_{++}, u_{--}) \geq 0.5$. Swap error occurs if $u_{++} \leq u_{+-}$ or $u_{--} \leq u_{-+}$, including ties. JointPairAcc requires PairAcc success without Swap error. We report pair-level percentages; same-answer counts use exact equality of parsed interval lists.

Table 1. SpotSound-Bench (400 queries, %). Baselines are from SpotSound Table 3 [8]; RelTwin reports three-seed means. Best and second-best scores are bold and underlined.

Method	mIoU $\uparrow$	R1@0.3 $\uparrow$	R1@0.5 $\uparrow$
<i>Task-specific grounding models</i>			
WTATG	38.4	53.0	34.5
AM-DETR	19.5	25.5	15.0
<i>Proprietary audio-language models</i>			
Gemini-2.5-Flash	25.7	34.8	27.8
Gemini-2.5-Pro	19.7	20.0	17.3
<i>General-purpose open audio-language models</i>			
Kimi-Audio	3.2	3.0	1.3
Qwen2-Audio	4.4	5.5	1.8
Audio Flamingo 3	7.6	8.8	2.8
<i>Temporal grounding-enhanced models</i>			
TimeAudio	10.2	7.8	1.8
SpotSound-Q	49.9	65.0	51.3
SpotSound-A	<u>57.9</u>	<u>76.5</u>	<u>59.5</u>
RelTwin (3 seeds)	59.16	77.17	62.08
Δ vs. prior main-table best	+1.26	+0.67	+2.58

Predicting the union of two disjoint, equal-length targets for both queries passes PairAcc (IoU 0.5) but fails JointPairAcc. Inspired by paired compositional evaluation [15], JointPairAcc distinguishes query-specific selection from coverage.

4. EXPERIMENTAL SETUP

Data and evaluation protocol. Adaptation uses 40 ESC-50 classes and 256 constructed recordings: 1,024 relation queries (512 pairs) and 512 rehearsal queries. The remaining 10 classes provide 80 recordings, 320 queries, and 160 inverse pairs for relation development. Training and development use disjoint classes and original Freesound IDs at the adaptation stage. The constructed split diagnoses query-window binding; public evaluation measures grounding performance on all 400 SpotSound queries over 387 audios. Both relation-development and public benchmark results informed method development.

Training settings. Both methods retain SpotSound-A’s LoRA adaptation [16] on Audio Flamingo 3, with rank 8, scaling parameter $\alpha = 16$, dropout 0.1, and approximately 20.19M trainable adapter parameters. SFT and RelTwin use 256 AdamW [17] updates, zero weight decay, learning rate 5×10^{-6} , gradient clipping 1.0, $\tau = 1$, and $\beta = 0.5$; RelTwin uses $\lambda = 1$. Each update processes one inverse-query pair (two relation queries) and one category-rehearsal query. Gradients accumulate over these components, with no accumulation across pairs. The paired comparison uses seed 0, shared initialization, example order, training code, and software environment to assess the effect of candidate supervision. Up-

dates are matched, but FLOPs and wall time are not: RelTwin first scores four query-answer combinations without gradients and then recomputes weighted gradients, whereas SFT backpropagates through the two positive answers. Both include rehearsal.

Inference. We evaluate paired SFT seed 0 and RelTwin seeds 0/1/2 after 256 updates. SpotSound-A and the adapted models share the inference protocol: 16 kHz mono audio, bfloat16 inference, and greedy decoding (one beam, no sampling), with at most 128 new tokens per query.

Comparison roles. Table 1 compares RelTwin with published SpotSound-Bench results; Table 2 separates the backbone reference, matched training comparison, and three-seed replication. The backbone scores 58.42 mIoU under our inference protocol, distinct from its published 57.9. RelTwin’s three-seed difference against this backbone reference is +0.75 points, with audio-group CI $[-0.24, 1.74]$ spanning zero. The +1.26 improvement in Table 1 is measured against the published main-table best.

Statistical analysis. We compute 95% confidence intervals from 20,000 bootstrap resamples, keeping queries from the same audio together. Relation comparisons additionally resample 39 connected groups of recordings sharing original sources (largest: 18 audios), keeping reused excerpts within each group. These intervals quantify sample uncertainty for the evaluated seeds, without correction for development selection.

5. RESULTS AND ANALYSIS

5.1. Timestamp contrasts recover query-specific selection

The paired diagnostic directly tests whether reversing a query changes the selected event window (Table 2). Paired SFT returns identical answers for 37 of 160 inverse-query pairs, spanning 22 recordings; RelTwin reduces this count to one. Of those 37 pairs, 27 become jointly correct. This recovery connects answer differentiation to the requested relation, rather than merely to producing different timestamps. Figure 1 shows an example: RelTwin selects the corresponding window for each query, whereas SFT repeats one answer.

JointPairAcc increases by 31.25 points over paired SFT, with source-group 95% CI $[22.35, 40.50]$, while Swap error decreases by 32.50 points. Across all pairs, 51 change from incorrect to jointly correct, one changes in the opposite direction, and 24 remain incorrect. These transitions show that the aggregate gain is driven by recovering query-specific answers. Relation mIoU also increases by 13.97 points (CI $[9.84, 17.51]$), connecting improved selection with temporal overlap. The example retains some boundary undercoverage, illustrating why relation selection and interval accuracy are measured separately.

Across RelTwin’s three training seeds, JointPairAcc averages 81.46 and identical-answer counts are 1, 0, and 0. Thus,

Table 2. Query–window binding on the synthetic relation-development set and transfer to SpotSound. Panel (b) compares seed-0 models with matched data and updates. Panel (c) reports means $\pm$ sample SD; same-answer counts follow seed order 0/1/2. Joint denotes JointPairAcc; same-answer counts identical timestamp answers to inverse queries. Scores are percentages except pair counts.

Method	SpotSound	Relation development (160 inverse-query pairs)			
	mIoU $\uparrow$	mIoU $\uparrow$	Joint $\uparrow$	Swap error $\downarrow$	Same-answer pairs $\downarrow$
<i>(a) Backbone reference: shared evaluation protocol</i>					
SpotSound-A	58.42	37.64	0.63	65.00	15/160
<i>(b) Matched training comparison: seed 0</i>					
Paired SFT	58.42	74.58	53.13	40.00	37/160
RelTwin	59.43	88.55	84.38	7.50	1/160
RelTwin – Paired SFT	+1.02	+13.97	+31.25	−32.50	−36 pairs
<i>(c) RelTwin replication: three training seeds</i>					
RelTwin	59.16 $\pm$ 0.26	87.05 $\pm$ 1.43	81.46 $\pm$ 2.53	8.13 $\pm$ 0.63	(1, 0, 0)/160

the reduction in same-answer collapse is reproduced across the three RelTwin runs, alongside the primary paired-SFT comparison.

5.2. Binding gains persist after timing intervention

The timing intervention tests whether layout-specific silence explains the binding advantage. We recompose all 80 development audios with 1.5 s lead-in, 0.25 s within-window gap, 3 s between windows, and 2 s tail, preserving event waveforms, order, and queries. Frozen SFT/RelTwin seed 0 models evaluate one template (160 queries). The 40 BA-first recordings receive the changed timing layout; the other 40 AB-first recordings remain waveform-identical.

On the 40 changed recordings, RelTwin achieves 87.50 JointPairAcc versus 67.50 for SFT: +20.00 points, source-group CI [2.94, 33.33]. Under the original layout, the corresponding scores are 85.00 and 42.50. The intervention improves SFT and narrows the gap from 42.50 to 20.00 points, showing that timing layout affects the comparison. RelTwin nevertheless retains an advantage when the silence pattern changes. Across all 160 queries, uniform-timing mIoU is 81.34 for SFT and 90.65 for RelTwin. Together, these post-development results show that the original silence pattern does not fully explain the query–window binding gain.

5.3. Public grounding performance

RelTwin attains the highest scores on all three metrics among the methods in Table 1. Across three training seeds, mIoU averages 59.16, exceeding the prior main-table best of 57.9 by 1.26 points. R1@0.3 and R1@0.5 improve by 0.67 and 2.58 points, respectively. The larger recall gain at IoU 0.5 shows that the improvement extends beyond loose overlap to predictions satisfying the stricter localization criterion. The strongest published comparator, SpotSound-A, uses the same

Audio Flamingo 3 backbone, placing these gains against a model already adapted for temporal grounding.

The matched training comparison in Table 2 complements these published-score comparisons. On the same 400 public queries, RelTwin improves over paired SFT by 1.02 mIoU points, with audio-group CI [0.23, 1.81]. The paired outcomes are 92 improvements, 239 ties, and 69 regressions, including two IoU decreases of at least 0.5. Public localization gains therefore accompany the recovery of query-specific selection observed in the controlled relation diagnostic.

5.4. Scope

Repeated excerpts and fixed templates enable controlled binding tests. Alternative hard-negative constructions, matched multi-seed replication, and natural overlapping or long-range relations remain open evaluations. Adaptation-stage source separation does not audit backbone pretraining overlap.

6. CONCLUSION

Event presence alone does not determine which interval matches a relation query. RelTwin trains query–window binding through locally valid timestamp contrasts: competing windows have real acoustic support, but their correctness depends on the query. Paired diagnosis and the timing intervention show improved joint localization over paired SFT, alongside gains on SpotSound-Bench. These results support explicit query-conditioned competition between timestamp answers while retaining direct timestamp generation at inference.

7. ACKNOWLEDGMENTS

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8. COMPLIANCE WITH ETHICAL STANDARDS

This computational study reuses public audio from ESC-50 [14] and SpotSound-Bench [8], including programmatically constructed sequences. No new recordings, participant recruitment, human or animal experiments, or listening studies were undertaken.

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